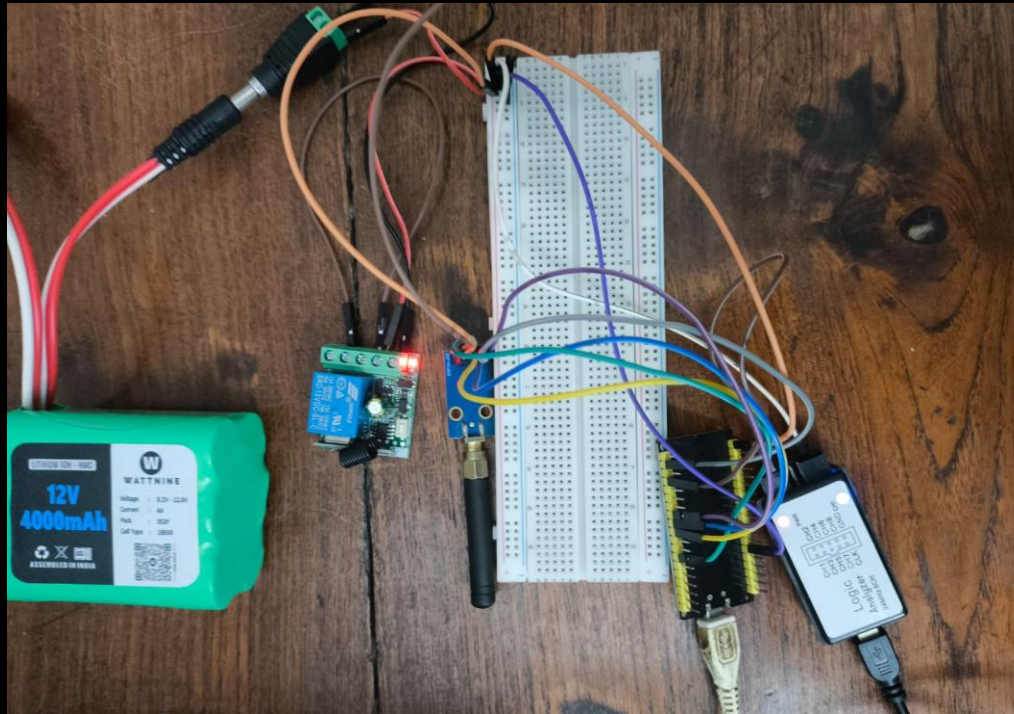
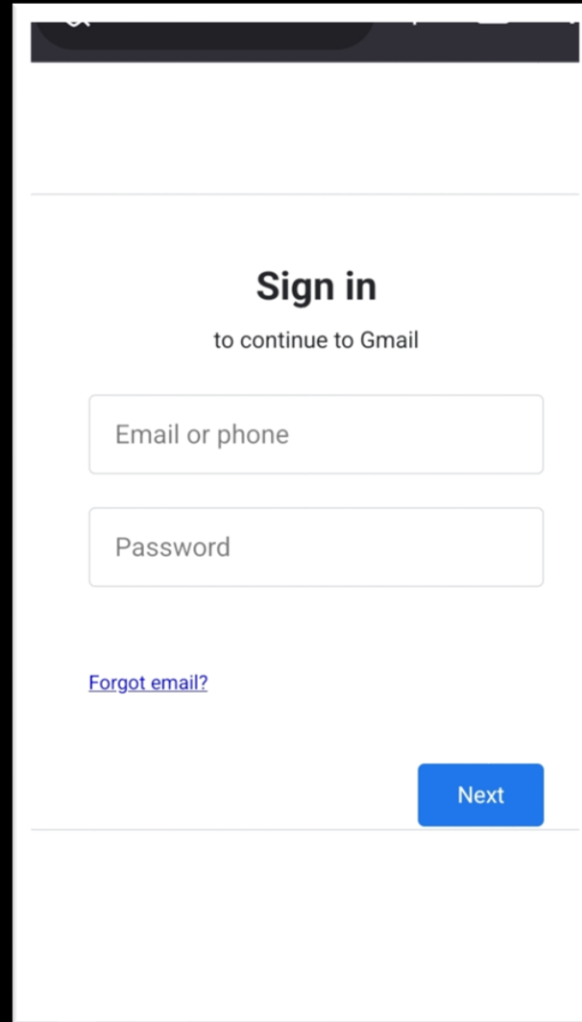


01/ Hardware Architecture & Research Setup



Isolated home lab testing environment ESP32, CC1101 sub-GHz transceiver, and analysis tools

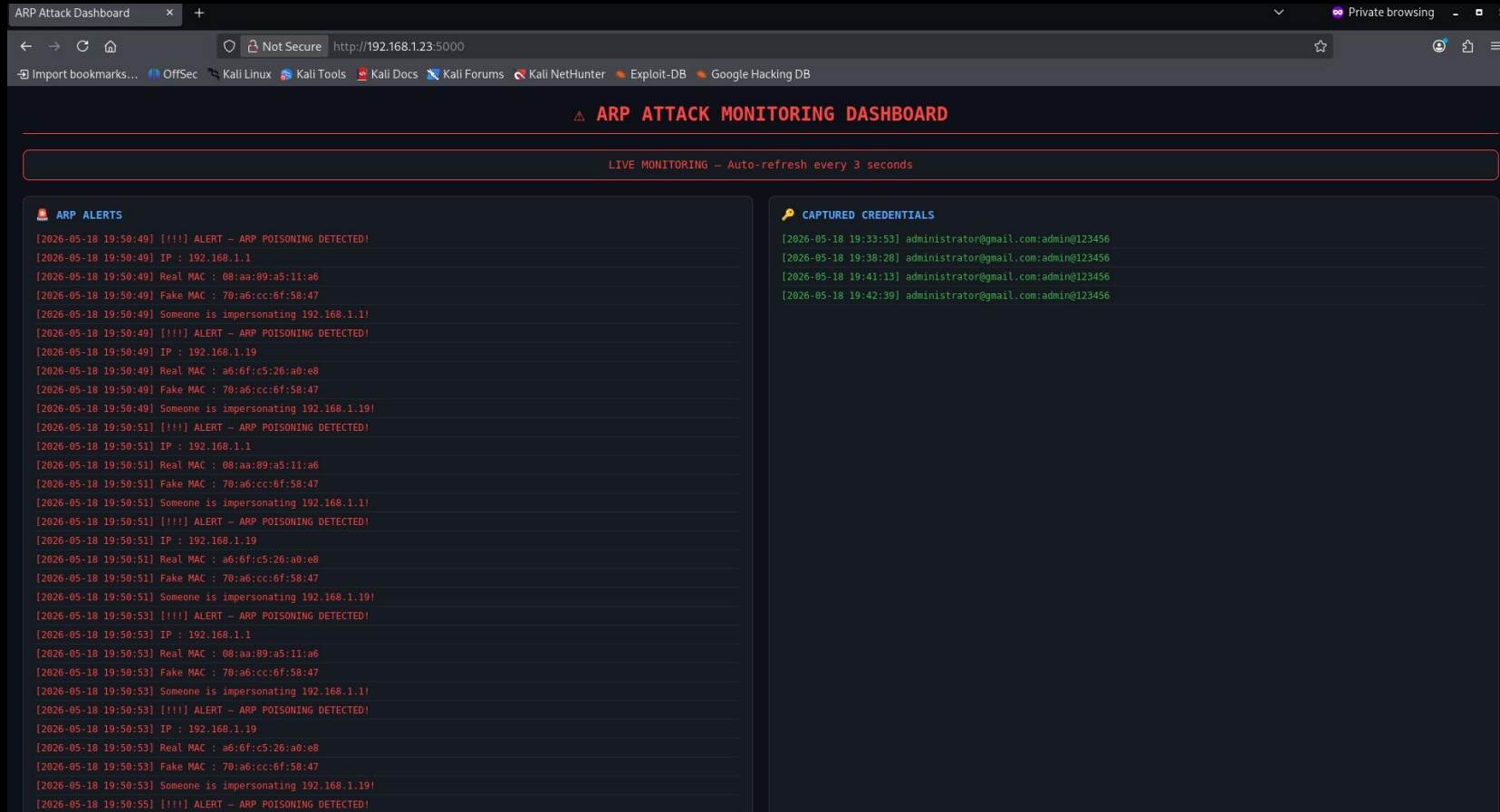
02/ Layer 7 Credential Capture Interface



The image shows a mobile device screen displaying a captive portal. At the top, there is a dark status bar. Below it, a horizontal line separates the header from the main content. The main content area has a white background and contains the following elements: the text "Sign in" in bold, followed by "to continue to Gmail" in a smaller font. Below this are two input fields: the first is labeled "Email or phone" and the second is labeled "Password". Under the "Password" field is a blue link that says "Forgot email?". At the bottom right of the form area is a blue button with the text "Next". A horizontal line is positioned below the "Next" button.

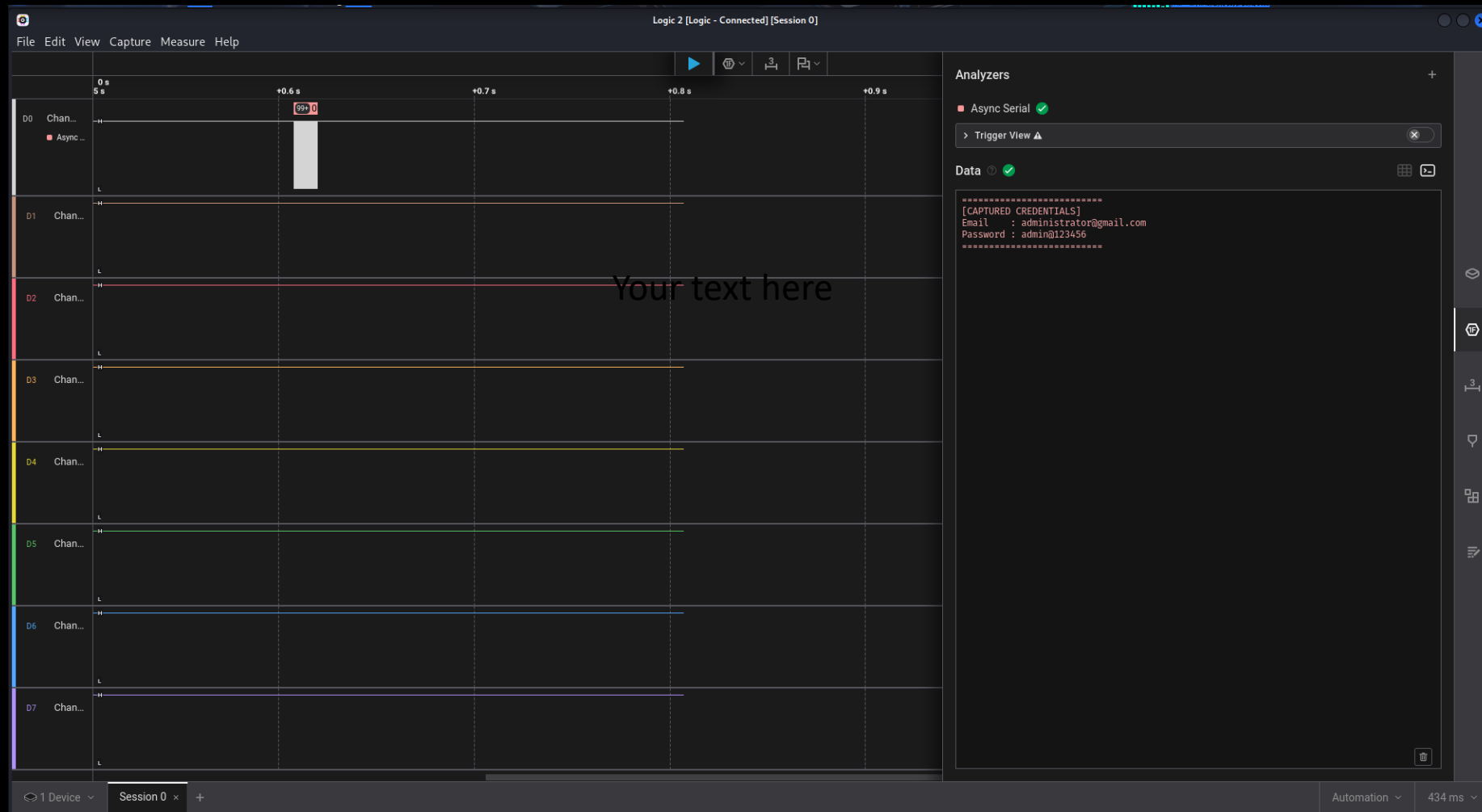
Captive portal served to the victim device following successful Layer 2 and Layer 3 interception

03/ Centralized Attack Monitoring Dashboard



Real time Flask dashboard that monitors network data and flags potential data leaks

04/ Physical Bus Auditing & Waveform Decoding



Decoding unencrypted UART/SPI serial bus communication lines using a Saleae Logic Analyzer to capture data transitions.

05/ Real-Time Detection

```
[kali@kali: ~]$ sudo python3 detector.py
[2026-05-18 17:58:27] [*] ARP Anomaly Detector started...
[2026-05-18 17:58:28] [*] Watching 4 trusted devices

[2026-05-18 17:58:35] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:35] IP : 192.168.1.1
[2026-05-18 17:58:35] Real MAC : 08:aa:89:a5:11:a6
[2026-05-18 17:58:35] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:35] Someone is impersonating 192.168.1.1!

[2026-05-18 17:58:35] [+] New device seen: 192.168.1.1 → 70:a6:cc:6f:58:47
[2026-05-18 17:58:35] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:35] IP : 192.168.1.19
[2026-05-18 17:58:35] Real MAC : a6:6f:c5:26:a0:e8
[2026-05-18 17:58:35] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:35] Someone is impersonating 192.168.1.19!

[2026-05-18 17:58:35] [+] New device seen: 192.168.1.19 → 70:a6:cc:6f:58:47
[2026-05-18 17:58:37] [+] New device seen: 192.168.1.23 → 70:a6:cc:6f:58:47
[2026-05-18 17:58:37] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:37] IP : 192.168.1.1
[2026-05-18 17:58:37] Real MAC : 08:aa:89:a5:11:a6
[2026-05-18 17:58:37] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:37] Someone is impersonating 192.168.1.1!

[2026-05-18 17:58:37] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:37] IP : 192.168.1.19
[2026-05-18 17:58:37] Real MAC : a6:6f:c5:26:a0:e8
[2026-05-18 17:58:37] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:37] Someone is impersonating 192.168.1.19!

[2026-05-18 17:58:39] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:39] IP : 192.168.1.1
[2026-05-18 17:58:39] Real MAC : 08:aa:89:a5:11:a6
[2026-05-18 17:58:39] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:39] Someone is impersonating 192.168.1.1!

[2026-05-18 17:58:39] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:39] IP : 192.168.1.19
[2026-05-18 17:58:39] Real MAC : a6:6f:c5:26:a0:e8
[2026-05-18 17:58:39] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:39] Someone is impersonating 192.168.1.19!

[2026-05-18 17:58:41] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:41] IP : 192.168.1.1
[2026-05-18 17:58:41] Real MAC : 08:aa:89:a5:11:a6
[2026-05-18 17:58:41] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:41] Someone is impersonating 192.168.1.1!

[2026-05-18 17:58:41] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:41] IP : 192.168.1.19
[2026-05-18 17:58:41] Real MAC : a6:6f:c5:26:a0:e8
[2026-05-18 17:58:41] Fake MAC : 70:a6:cc:6f:58:47
[2026-05-18 17:58:41] Someone is impersonating 192.168.1.19!

[2026-05-18 17:58:43] [!!!] ALERT - ARP POISONING DETECTED!
[2026-05-18 17:58:43] IP : 192.168.1.1
```

Python anomaly detection engine catching unauthorized MAC
modifications